2017-2018 – DEEQA – TSE – Advanced Macroeconomics

Lecture 2 : Revisiting the Evidence for Cycles

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Disclaimer

These are the slides we are using in class. They are not self-contained, do not always constitute original material and do contain some "cut and paste" pieces from various sources that we are not always explicitly referring to (not on purpose but because it takes time). Therefore, they are not intended to be used outside of the course or to be distributed. Thank you for signaling me typos or mistakes at f.portier@ucl.ac.uk.

Note

- Based mainly on my work with Paul Beaudry & Dana Galizia
- Also based of work of Paul with Meynging Wei & Natasha Kang

Introduction

- This lecture will be empirical with the goal of re-examining some basic questions regarding how we view/frame our discuss about macroeconomic fluctuations (see Lecture 1).
- Questions addressed
  - × What are the main features that business cycle theory should be explaining ?
  - × Could we have been missing one really important element : *cyclicality* ?
  - × How does this relate to the financial cycle ?

- This requires return to the question : what do we mean by a cycl ?
- This has been the focus of Lecture 1.
- At least two different notions
  - A cycle could mean a tendency for a phenomena, such as a boom and bust (whether real or financial), to recur with a certain degree of regularity
  - Could alternatively mean : Co-movement features of many variables at a given frequency range
- The later is generally implied for the business cycle (as we have seen in Lecture 1)...
- ...while the former seems to be the current focus for the financial cycle debate (e.g., BIS reports).

- Usually viewed a subset of macroeconomic fluctuations that arises between 6 and 32 quarters
- At these frequencies, generally no marked cycle found– in terms of regular recurrence of boom and bust
  - Regular occurence should be seen noticed as a peak in the spectral density .
- However, at these frequencies, instead very strong co-movement between many variables. This has become the defining feature. And this substantially influences theories.

- Much less agreement of what it refers to. What frequency are we talking about ? Is it about co-movement or regular recurrence ?
- One of the difficulty that plagues this issue (which is also shared by business cycle theory), how to decompose trend and cycle.
  - For non-stationary variable this is very difficult. Reason : filtering data (removing trend) can spuriously create cycles.
  - For stationary variables, this can be easier. But these may be rare.
- Some studies (ex : BIS) suggest that there is a financial cycle (recurrent), but it is at a lower frequency than the business cycle.

- Explore anew the notion of business cycles and financial cycles and their link
- Take a slightly wider notion of business cycles and focus initially on variables that are close to stationary or plausibly stationary (to avoid trend-cycle decompositions).
- Main claim
    1. There is much more evidence of cyclical behaviour than commonly recognized, in the sense of recurrent forces
    2. In particular, real and financial variables appear to share similar recurrent cyclical features at periodicity of about 8 to 10 years.
    3. This common cycle is only slightly longer than traditional BC definition, but shorter than current suggestion for the financial cycle. Believe this is still business cycle analysis, not some other medium run phenomena.
- Today concentrate on U.S. data, but we have been exploring other countries.

- Recurrent cyclical properties does not refer to some type of deterministic forces.
- The idea that a boom may sow the seeds of a subsequent bust does not mean this arises in a deterministic fashion
- It could for example represent a stochastic version of the old fashion Samuelson multiplier-accelerator model, where the accelerated is very strong and the system exhibits large complex roots.
- Conventional wisdom : *"Typical Spectral Shape of an Economic Variable"* (GRANGER 1969)

FIGURE 1 – GRANGER 1969

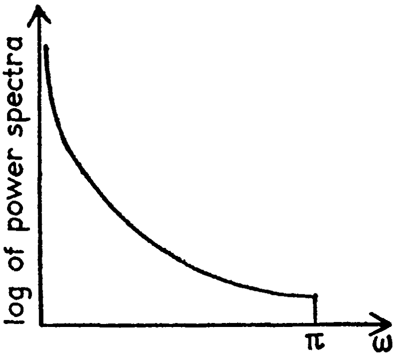
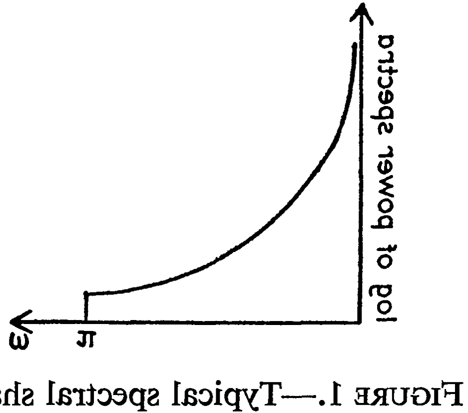


FIGURE 1.—Typical spectral shape.

Introduction

Conventional wisdom

FIGURE 2 – GRANGER 1969, Flipping the x axis



- $\rightsquigarrow$ look like a persistent AR(1) spectrum
- This suggested that business cycle theory should not be about cycles, it should be about co-movements (see SARGENT's textbook)
- Fluctuations are then modelled as coming from shocks that push the economy away from an otherwise stable steady state.
- I want here to question this conventional wisdom

1. Stationary Variables
2. Non Stationary variables

- We will repeatedly exploit spectral representation of the data.
- Why ?
- Can be informative for the question at hand
- This requires a few reminders....

- As seen in Lecture 1, spectral density is simply a transformation, a weighted sum, of the auto-covariance function.
- This is particular interesting when we want to think of a variable as being the sum of two (or more) independant components. Ex :

$$X_t = cyc_t + (lowfreq)_t$$

- and we have reason to believe that the mass of their variance arises at different frequencies. Why ?

$$S_x(\omega) = s_{cyc}(\omega) + s_{lf}(\omega)$$

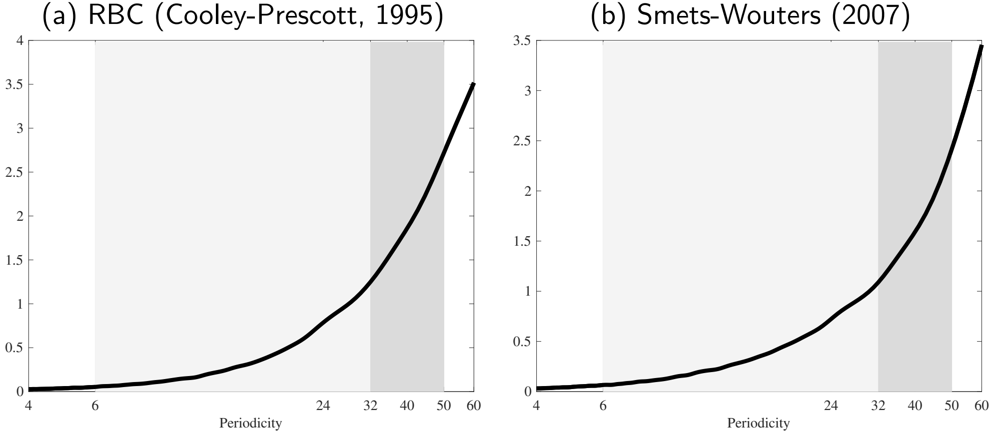
- we may be able to see the spectral of each component side by side.
- Allows to look for cyclical regularities by looking for peak in a given frequency range .

- Look at business cycle first
- Spend time on the behaviour of hours worked per capita
  - Potential nice example of stationary series with potentially both a business cycle component and a low frequency component
  - Try to clarify the attractive features of looking at spectrum
  - compare different treatment of the data
- Look at other stationary variables : ex : un-employment, capacity utilization
- Look at completely different cut of the data : exploiting only recession info.
- Look at financial cycle
- Look at longer series
- Look at trending/non stationary variables : Y, I, TFP...

- Begin by looking at the behaviour of hours worked.
- Let's preview by looking at implications from models

1. Stationary Variables

FIGURE 3 – Spectrum of Hours, Standard Models



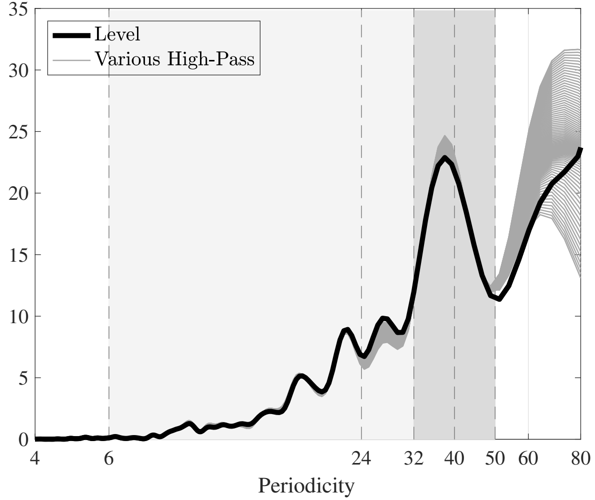
- This shape should not be too surprising. In RBC context, with RW technology, employment become a function of $\frac{K_t}{\Theta_t}$ and $\frac{K_t}{\Theta_t}$ is an AR(1).

$$l_t = \alpha \frac{K_t}{\Theta_t}$$

$$\frac{K_t}{\Theta_t} = \rho \frac{K_{t-1}}{\Theta_{t-1}} + \epsilon_t$$

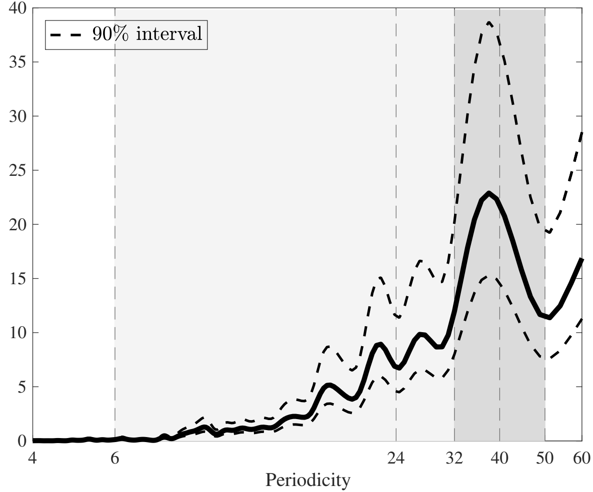
- These models do not produce "cyclical" behaviour (any regular recurrence) , which is general thought to be in line with data.
- Note that the spectral densities properties of l_t over the BC are not sensitive to the exact definition of frequencies associated with BC

FIGURE 4 – Spectrum : Non-Farm Business Hours Per Capita



1. Stationary Variables

FIGURE 5 – Spectrum : Non-Farm Business Hours Per Capita (with confindence bands)



- Is thinking about BC fluctuation as being associated with frequencies (periodicity) below 32 quarters appropriate ?
- Or should we think a bit longer ?

	PEAK-PEAK	QUARTERS
4)	**1960.II-1969.IV**	**39**
6)	*1973.IV-1980.I*	*25*
8)	**1981.III-1990.III**	**36**
9)	**1990.III-2001.I**	**43**
10)	2001.I-2007.IV	27
11)	**2007.IV-**	$>$**38**

1. Stationary Variables

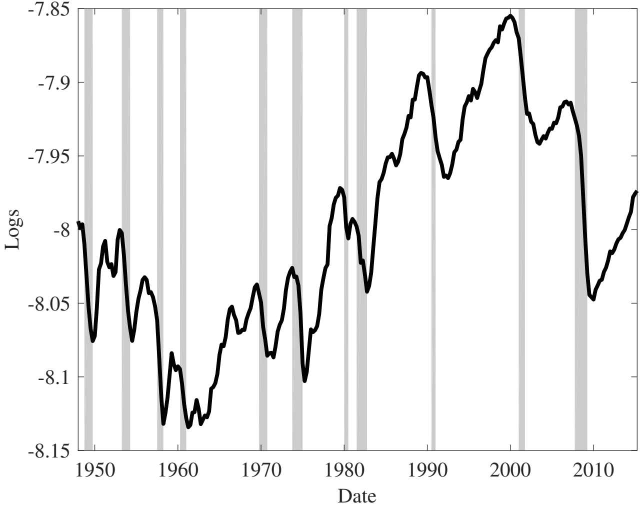
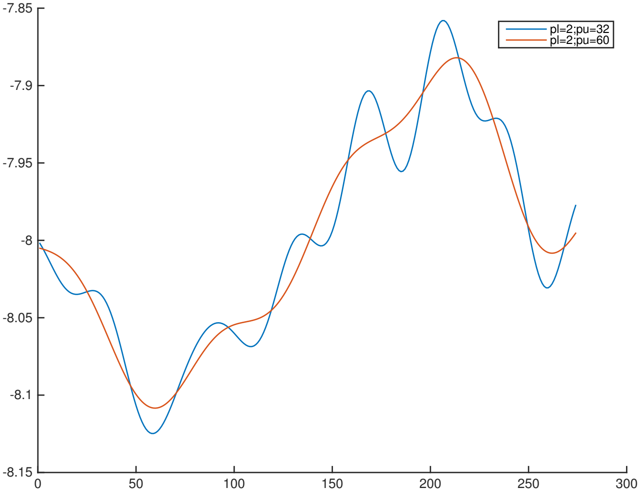


FIGURE 6 – Non-Farm Business Hours Per Capita

FIGURE 7 – Trends with 6-32 and 32-50 High-Pass Filters



- Could that hump in spectral density be spurious ?
- Could an AR(1) give rise to such a hump ?
- Take the best fitting AR(1)
- Draw 1000 draws of sample length
- Only detect humps similar to that observed in the data 1% of the simulations

- I said before that there is the same information in spectrum and autocorrelation function.
- So how do we detect a cycle by looking at the autocorrelation function ?

FIGURE 8 – Autocorrelation of NFB Hours per Capita

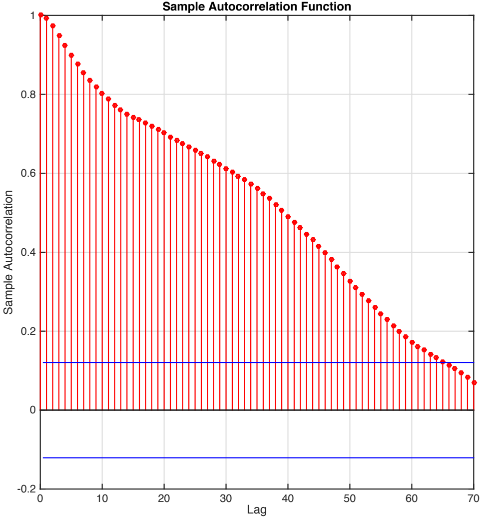
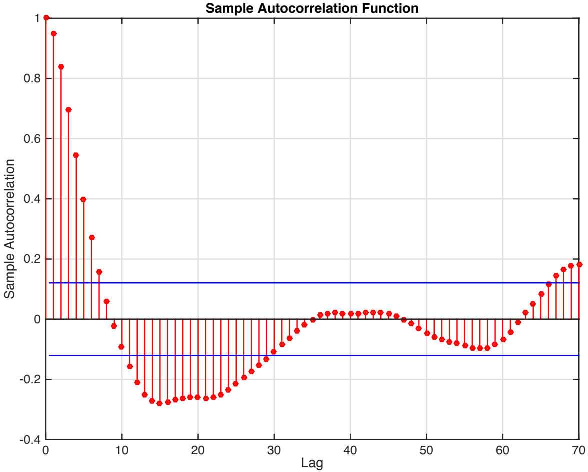
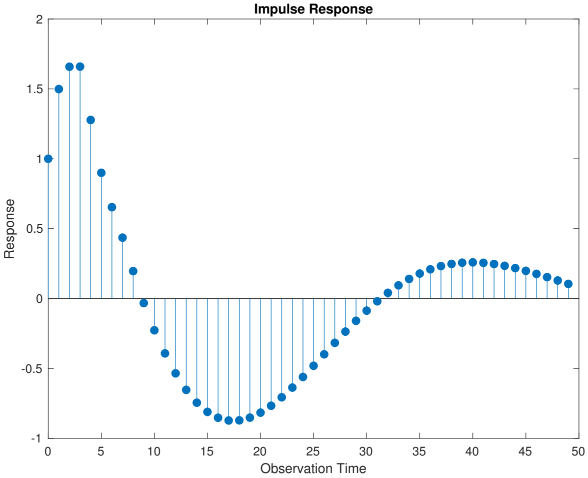


FIGURE 9 – Autocorrelation of NFB Hours per Capita, Filtered with High-Pass (100)



- Once the low frequencies in Hours are removed, we do see cycles in the auto-covariance.
- Let's estimate an ARMA on High-Pass(100) filtered Hours, and compute impulse response to a shock

FIGURE 10 – Impulse response associated with ARMA on High-Pass(100) Filtered Hours

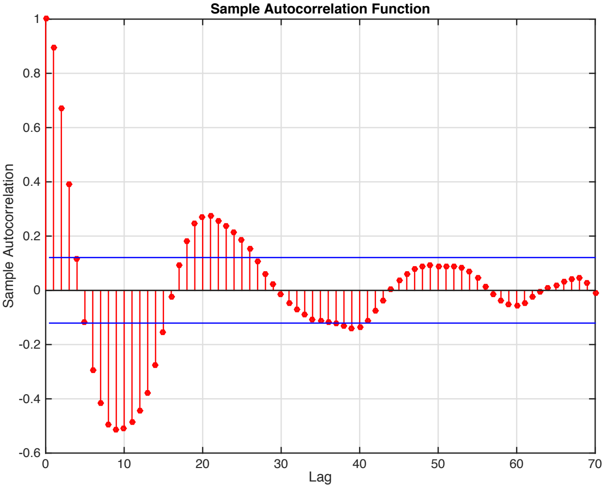


Filtering and spurious cycles

- Filtering will tend to create cycles – even when they don't exist– of a length similar (but slightly shorter) than the BP filter.
- Example High-Pass (32)

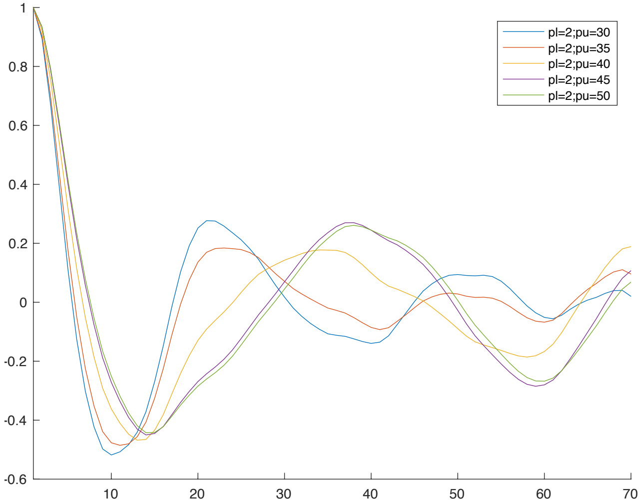
1. Stationary Variables

FIGURE 11 – Autocorrelation of NFB Hours per Capita, Filtered with High-Pass(32)



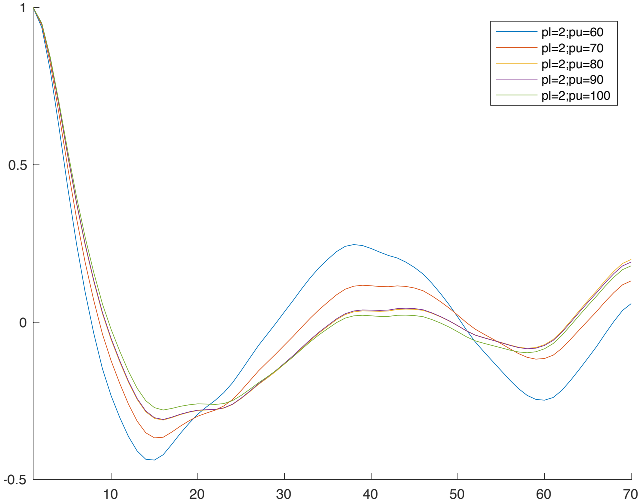
1. Stationary Variables

Figure 12 – Autocorrelation of NFB Hours per Capita, Filtered with High-Pass(Z)



1. Stationary Variables

Figure 13 – Autocorrelation of NFB Hours per Capita, Filtered with High-Pass(Z)



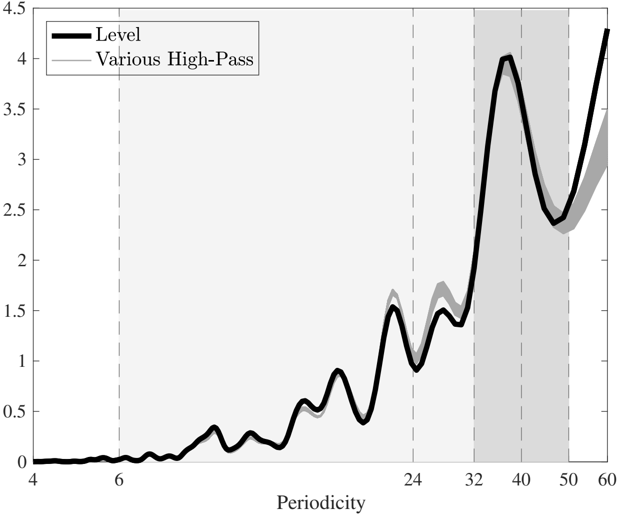
1. Stationary Variables

Other plausibly stationary activity variables

- Unemployment
- Capacity utilization

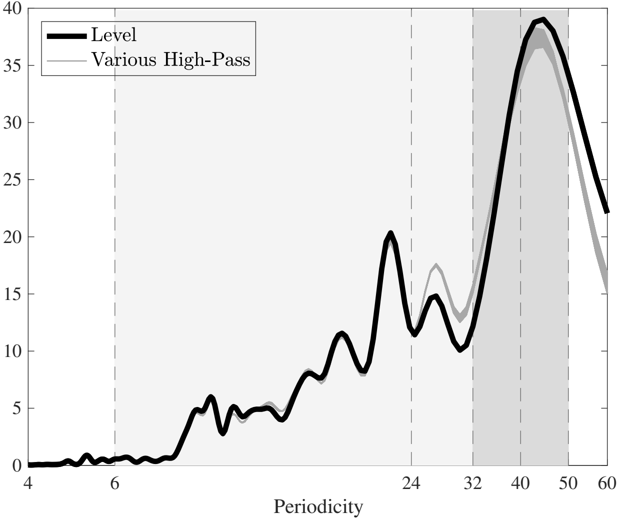
1. Stationary Variables

FIGURE 14 – Unemployment Rate



1. Stationary Variables

FIGURE 15 – Capacity Utilization Rate



- Time varying probability of recession
- This exploits only the time variation of the dummy that is 1 is the economy is in expansion and 0 iof it is in a recession.

FIGURE 16 – Probability of a recession *n* periods after a recession (short window)

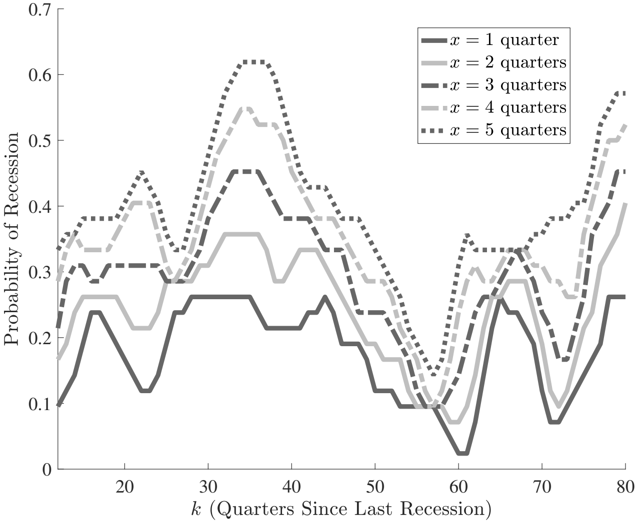
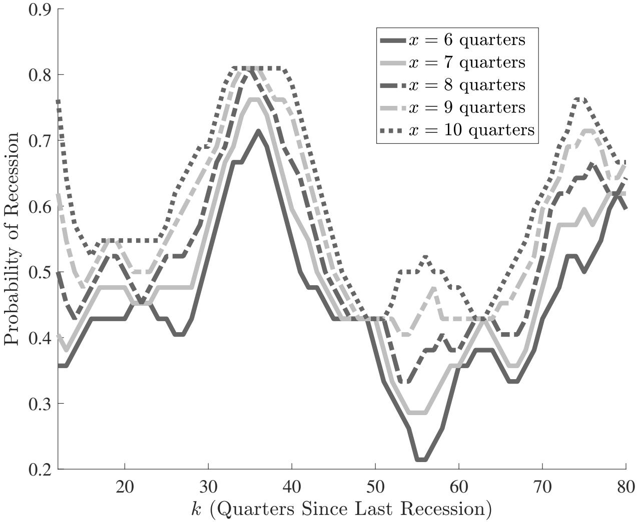


FIGURE 17 – Probability of a recession *n* periods after a recession (long window)

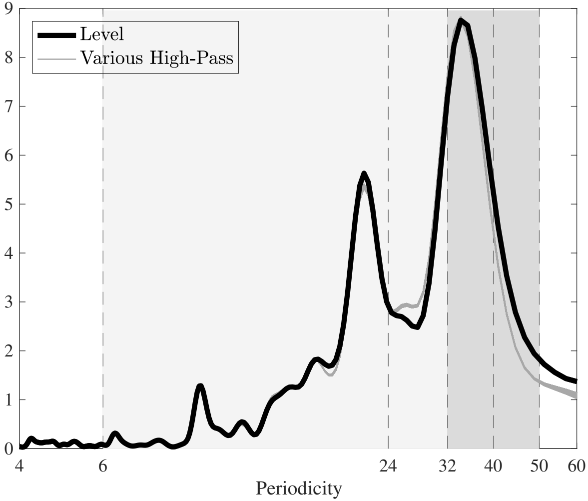


Some plausibly stationary financial variables

- spread, delinquency rates, financial condition indices

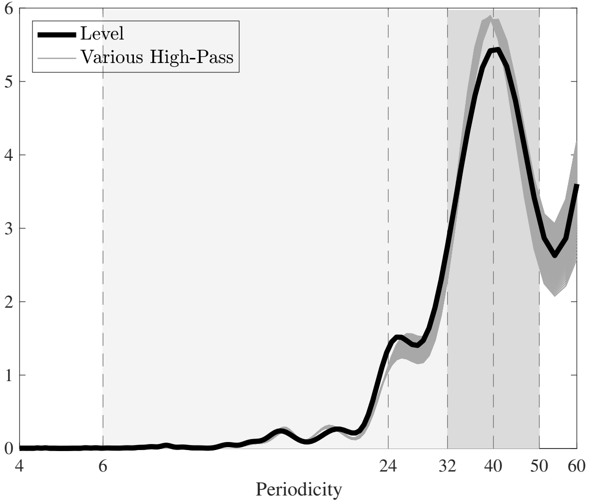
1. Stationary Variables

FIGURE 18 – Interest spread



1. Stationary Variables

FIGURE 19 – Delinquency Rate



1. Stationary Variables

FIGURE 20 – National Financial Conditions Index (Chicago Fed)

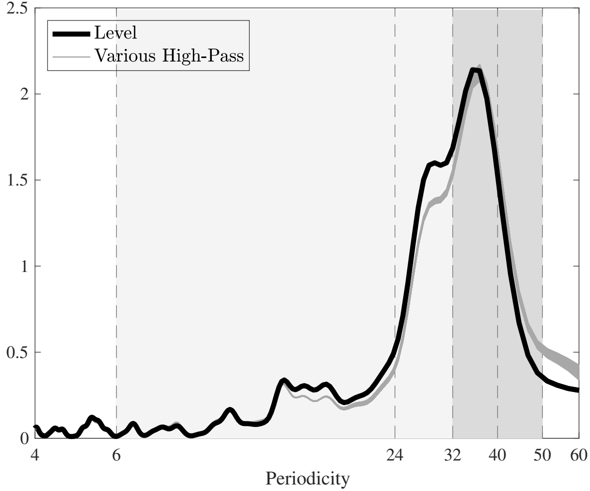
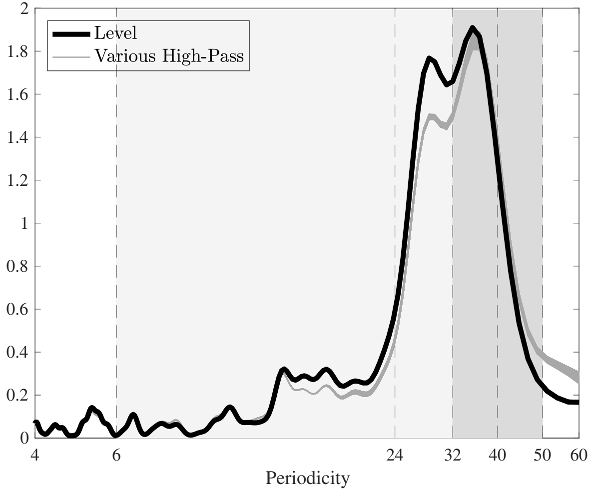


FIGURE 21 – National Financial Conditions Risk Subindex (Chicago Fed)

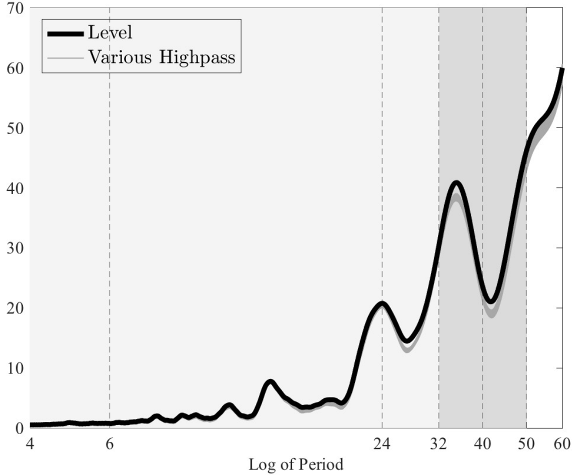


More Observations

- Some Historical Observations

1. Stationary Variables

FIGURE 22 – Unemployment since 1890



1. Stationary Variables

FIGURE 23 – Probability of a recession n periods after a recession (long window), since 1871

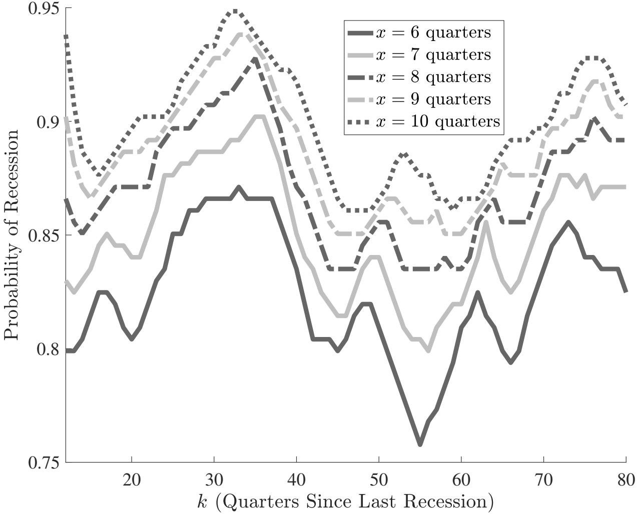


FIGURE 24 – Probability of a recession n periods after a recession (long window), since 1871

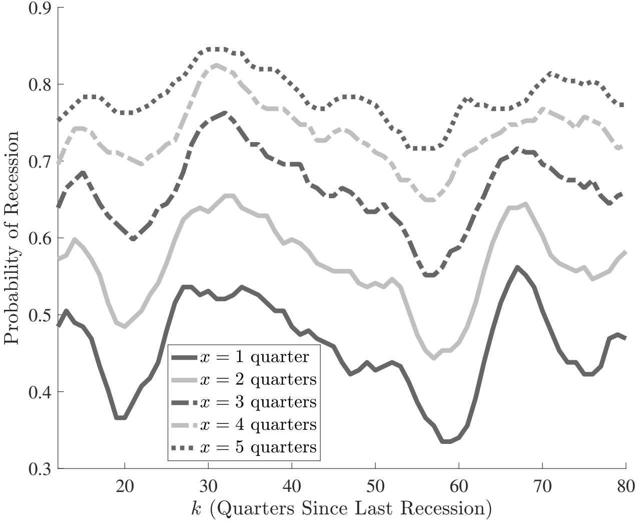
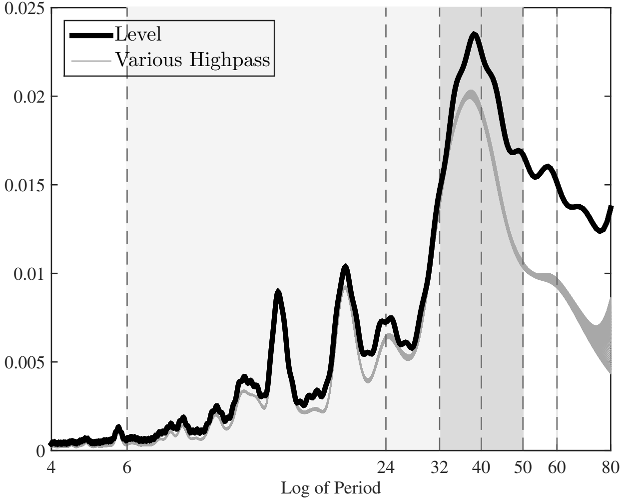
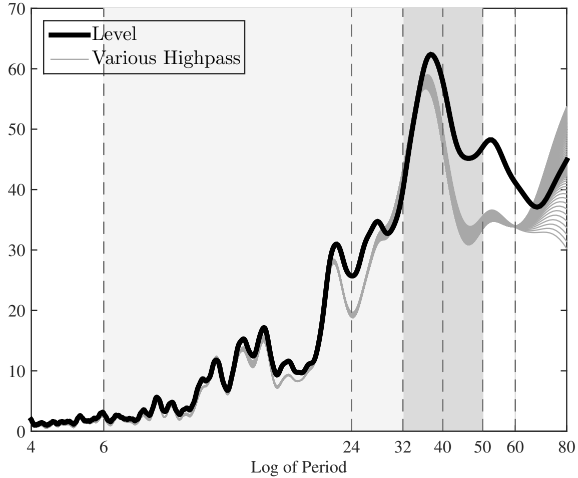


FIGURE 25 – Dividend Yield since 1871



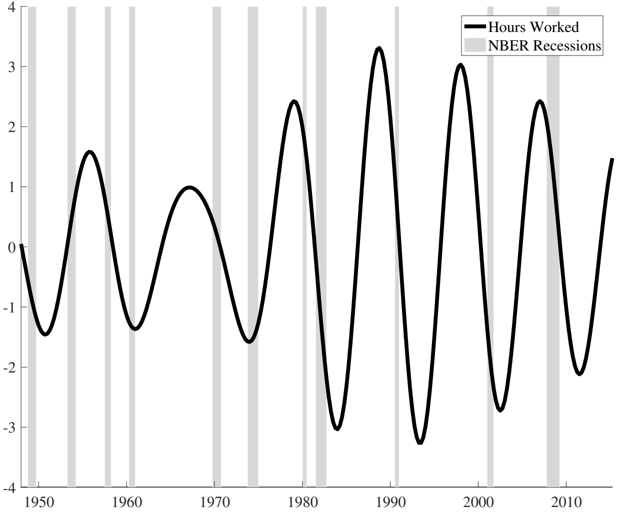
1. Stationary Variables

FIGURE 26 – BAA -10 year treasury spread since 1919



- . How can there still be sign of 8-10 year cycles in longer data, when recession where more common in the past
- May need to consider a more complex link between business cycles and recessions.
- Example : There may be business cycle forces that favour recurrent boom and busts.
  - During a boom phase (accumulation phase), the economy is robust to shocks and not much subject to recessions
  - However during a bust stage, the economy fragile and smaller shocks can make output fall. Therefore, it is a likely time for recessions, can could observe more than one : double dip.
  - Look at some evidence along this dimension.

Figure 27 – NFB Hours per Capita, 32-50 Quarters Component



- May ask, why are we finding so much evidence for cyclical behaviour in comparison to the literature ?
  - × Traditional (ex : Granger) : looks mainly at linearly de-trended trending variables.
  - × We focus more on stationary variables, where issue of trend-cycle decomposition is less of an issue.
  - × We use a loner sample
  - × We use padding technique (higher resolution technique) : which corresponds to simply adding zero to a series to allow more points before smoothing.

- Want to look at many other series, but most are trending and appear non-stationary.
- How to treat non-stationary series. No easy answer.
- Assuming variable is sum of stationary business cycle component plus non-stationary I(1) component, can look at spectrum of first difference.
- However, the first difference filter has the effect of emphasizing high frequency movement of the stationary component, plus the non stationary component in laid on top. So may be hard to see property of the stationary component.
- If ready to assume that non-stationary part is RW, we can (often) re-construct the spectrum for the stationary component.
- Can also put through BP filter, example 2-100 or 2-200. But not clear what this gives you. Can work well under certain circumstances.

- If

$$X_t = cyc_t + I(1)$$

- spectrum of first difference

$$S(\omega) = S_{\Delta cyc}(\omega) + S_{\Delta I(1)}(\omega)$$

- where $S_{\Delta cyc}(\omega)$ emphasizes high frequency movements, and $S_{\Delta I(1)}(\omega)$ may involve mass everywhere
- So may be difficult to infer property of the stationary component.
- In practice :
  1. take first difference,
  2. estimate the spectrum
  3. remove the zero frequency component
  4. invert the first difference filter to recuperate spectrum of the stationary part of the series

FIGURE 28 – Transfer of the First Difference Filter

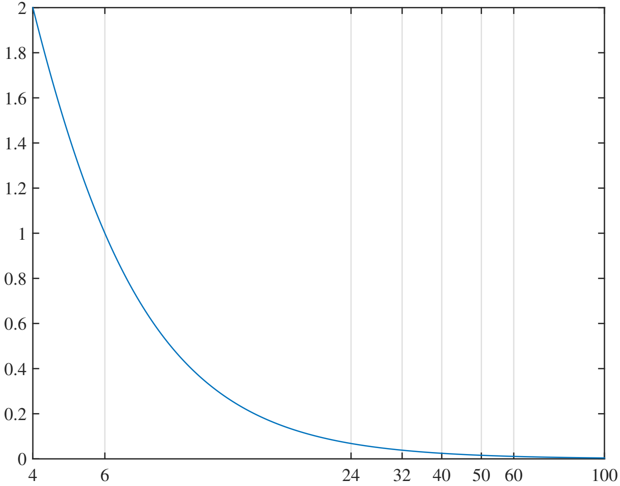
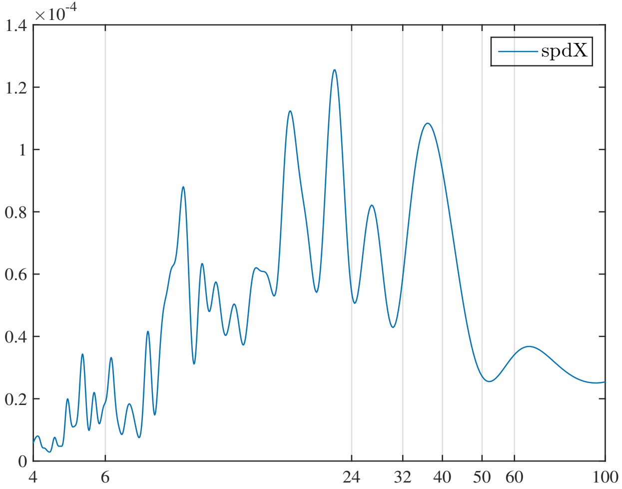
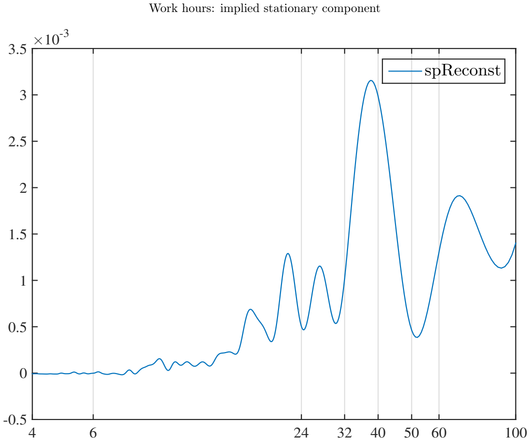


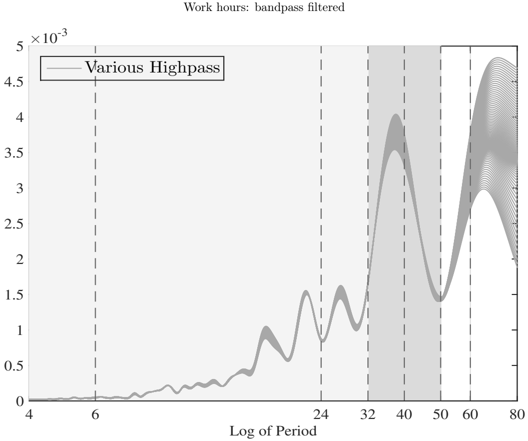
FIGURE 29 – Spectrum of first difference of hours worked



2. Non Stationary Variables

Figure 30 – Hours : RW Implied Component, BP filter

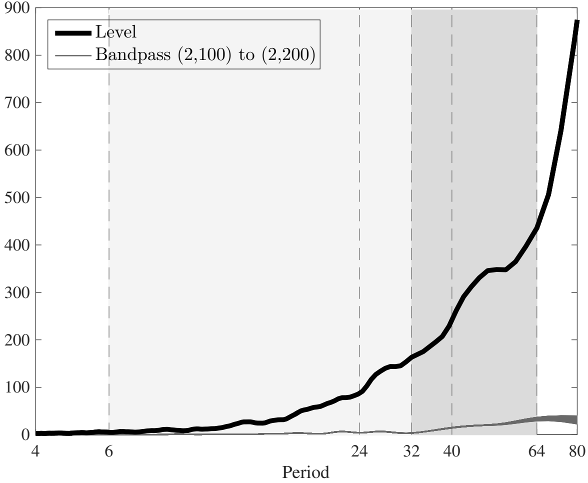




Some non-stationary activity variables

- GDP
- Investment

FIGURE 31 – Spectral Density, Level and High-Pass (2,100) to (2,200), Full Sample - Output



2. Non Stationary Variables

Figure 32 – Spectral Density, High-Pass (2,100) to (2,200), Full Sample - Output

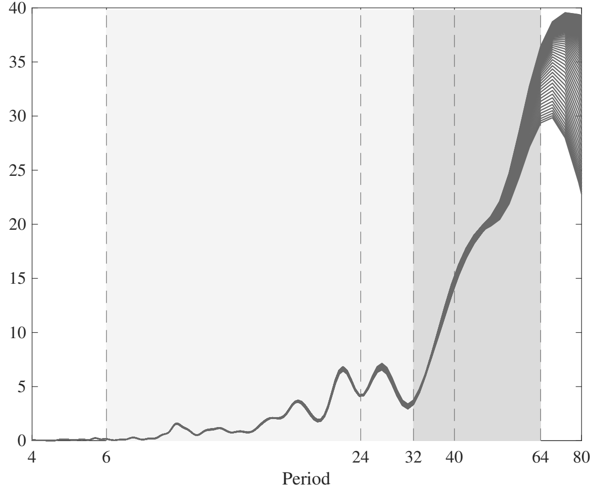
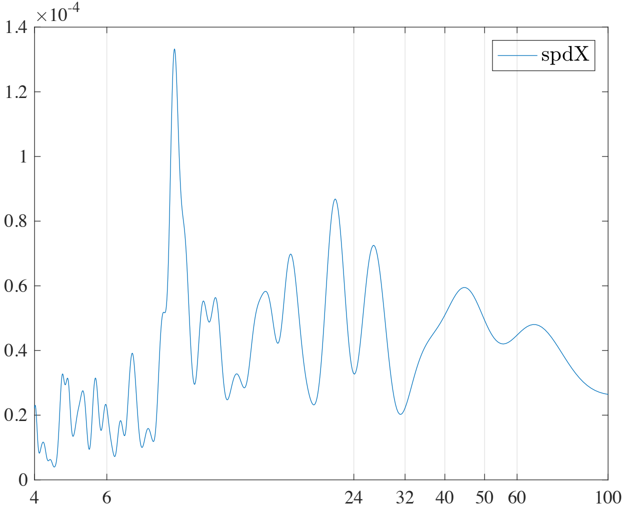


Figure 33 – GDP per capita : First difference



2. Non Stationary Variables

Figure 34 – Real GDP per capital : Implied Component and Various High-pass Filters

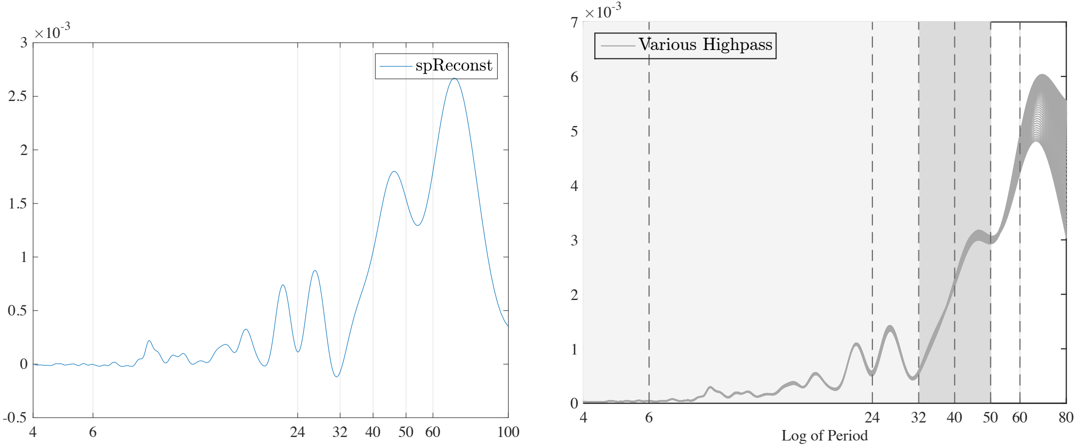


FIGURE 35 – Fixed Investment : First difference

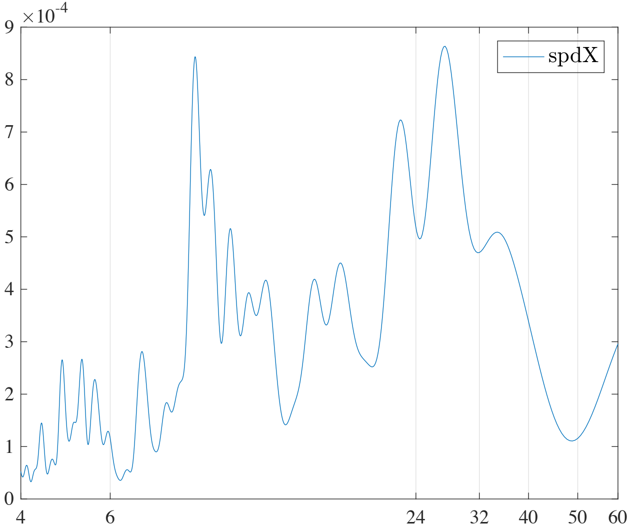


Figure 36 – Fixed Investment : Implied Component and Various High-pass Filters

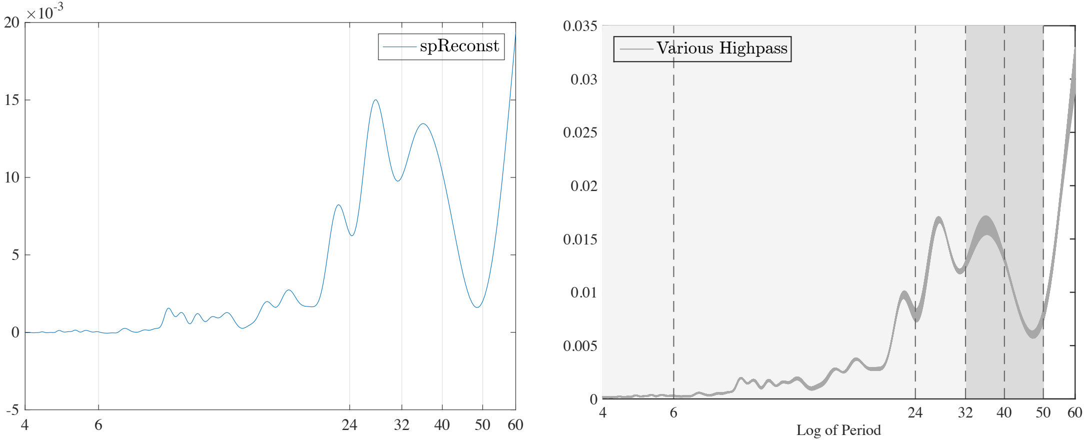
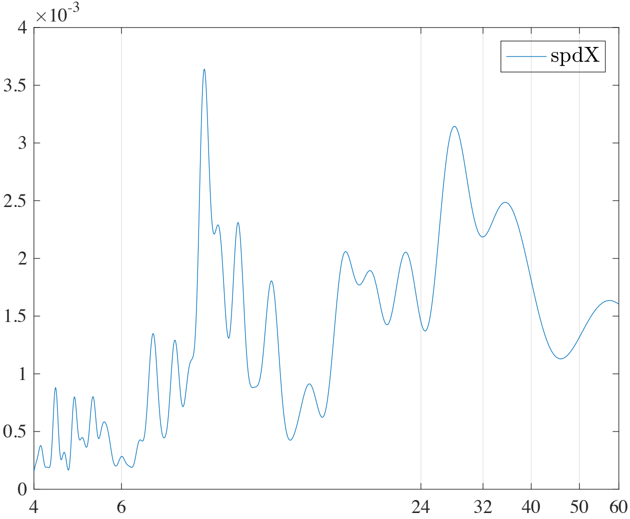


FIGURE 37 – Residential Inv : First difference



2. Non Stationary Variables

FIGURE 38 – Residential : Implied Component and Various High-pass Filters

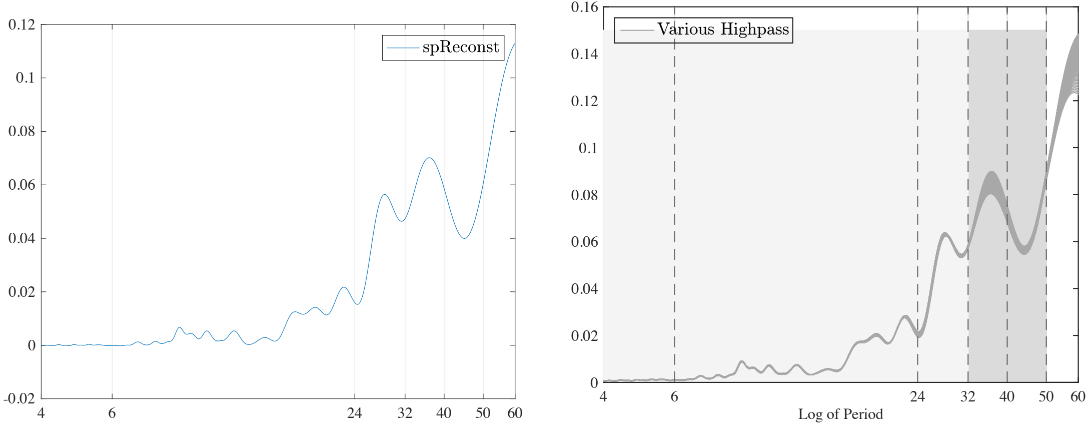
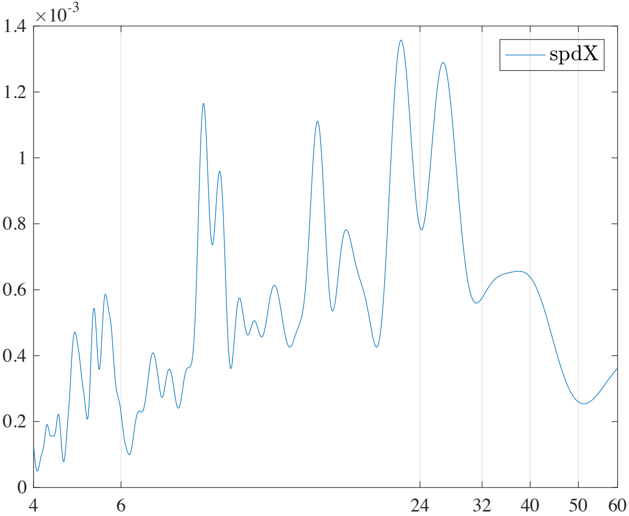
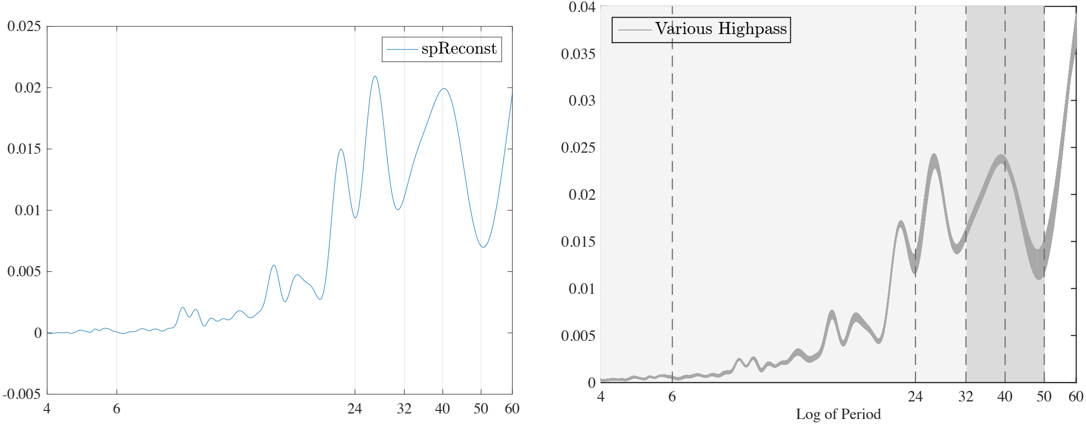


FIGURE 39 – Equipment Inv : First difference



2. Non Stationary Variables

FIGURE 40 – Equipment : Implied Component and Various High-pass Filters



- Look at TFP

2. Non Stationary Variables

Figure 41 – TFP : First difference

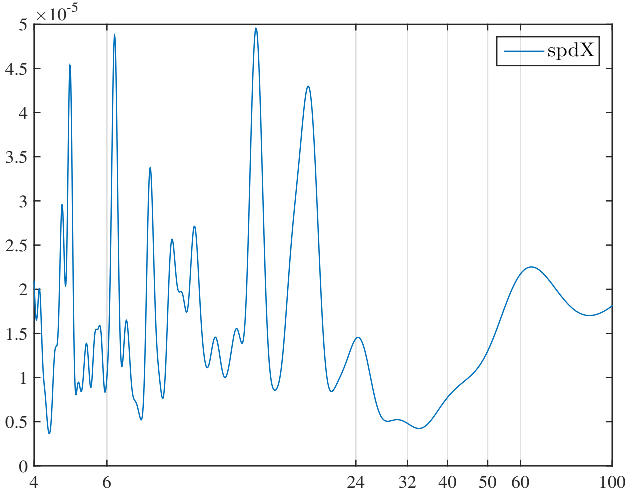
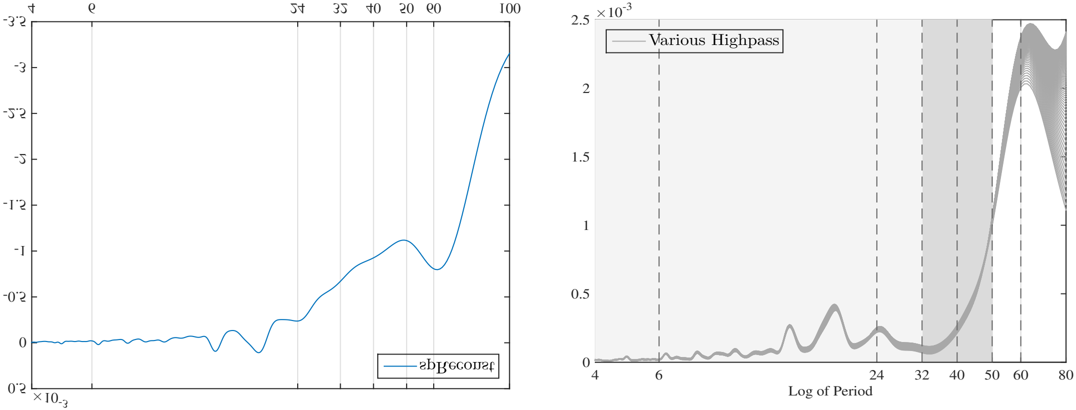
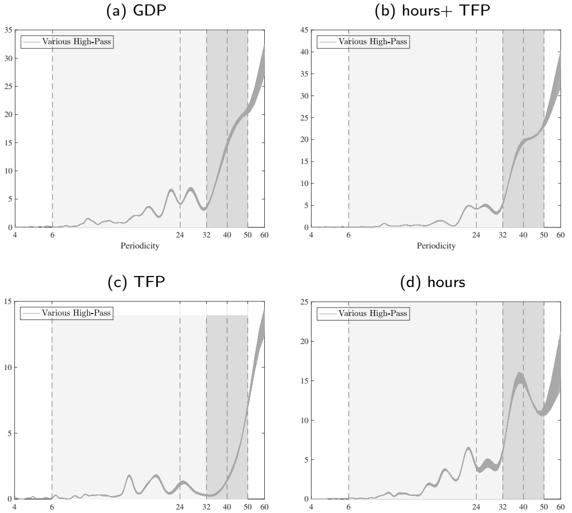


FIGURE 42 – TFP : Implied Component and Various High-pass Filters



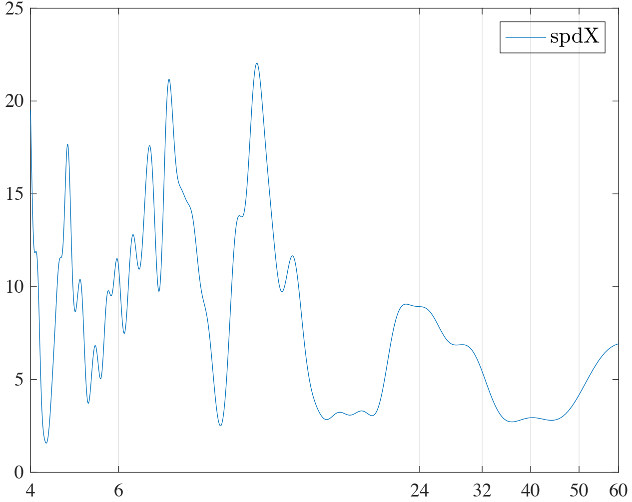
2. Non Stationary Variables

FIGURE 43 – Filtered GDP per capita



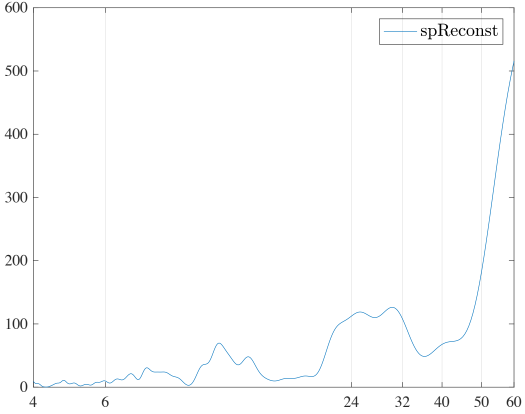
- Look at CPI inflation and relative price of energy (oil)

Figure 44 – Relative price of oil : First difference



2. Non Stationary Variables

FIGURE 45 – Reative price of oil : Implied Component, BP filter



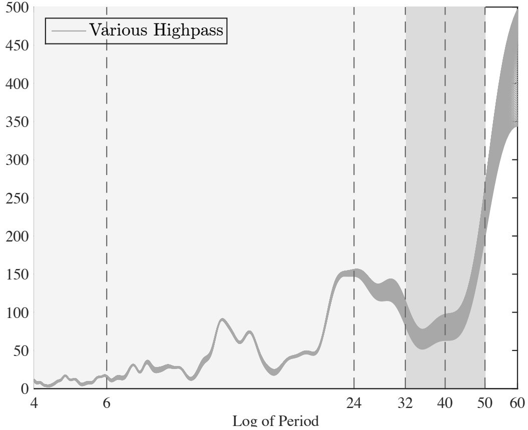
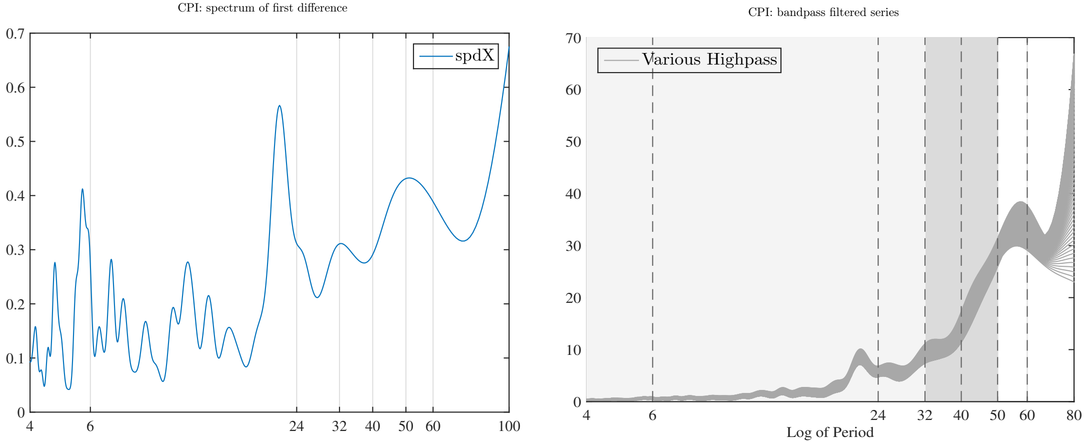


FIGURE 46 – CPI Inflation : First Difference, BP filter



Take away

- Seems warranted to consider BC fluctuations as applying to frequencies at least up to 40+ quarters
- Given this outlook, substantial evidence that there are real recurrent-cyclical forces in BC fluctuations
- Moreover, these fluctuation appeared shared by financial conditions
- The business cycle and the financial cycle may be very closely related and are cyclical
- Do not seem to coincide with TFP movement, inflation, energy

- What do observations of spectral peaks rule out.
    1. Difficult to reconcile with several business cycle models (e.g., RBC, Smets-Wouters)
    2. More generally, rule out for a variable x mechanisms of the type (or close to)

$$x_t = Bx_{t-1} + A_t$$

$$A_t = \rho A_{t-1} + \epsilon_t$$

    3. as this implies an AR(2) with real roots, which don't produce humps (need complex roots)
- need either more endogenous theory of BC, or a outside driver that itself is cyclical around 10 years

- Suggest a need for theories which generate substantial endogenous recurrent cyclical behaviour, which may include real financial linkages.
  - Ex : Kiyotaki-Moore (1997)
  - Beaudry, Galizia and Potier (2016, 2017)
- Question : Why are these not pick up in current estimation approaches
  - May not place enough weight on picking up this feature of the data.

Conclusion

- Business cycle analysis may be missing something extremely important : tendency of the real and financial system to exhibit (stochastically) recurrent boom bust phenomena at periodicities between 8 to 10 years.
- On many all fronts, common approaches may have been biased against recognizing these features. Taking unemployment instead of output as our key business cycle guide would be a good start.
- The presence of such cyclical behaviour suggests– to us – a need for a very strong (but somewhat slow moving) endogenous mechanism, which are not present in most models. May help explain the recurrent slow recoveries.
- Current popular avenues in BC research , which involve emphasis on idiosyncratic behaviour and shocks, may not be much help much here. What we may need to explain is unobserved homogeneity instead of unobserved heterogeneity.